

Statement of Objectives (SOO)

AEDC-Arnold AFB: SL Noise Reduction

1. Program Objectives

It is the intent of AEDC- Arnold AFB to conduct a comprehensive noise assessment study of adjacent buildings and areas, both outdoors and indoors, and to identify and propose viable solutions to mitigate noise levels and ensure compliance with applicable safety and health standards.

2. Contract Objectives

At a minimum, the contract shall define the following:

2.1. Work to-be Performed

Provide specific, performance-based, quantitative terms of the work to-be-done, in the form of a Statement of Work (SOW), to satisfy all project requirements established in SOO Attachment 1 – Performance Parameters. Follow on tasks are as follows:

2.1.1. Noise Assessment: Perform thorough noise level measurements in the Affected facilities outdoor areas in accordance with OSHA 1910.95 (12-hour shift, slow response) and in the indoor areas in accordance with ANSI/ASA S12.2-2019. This assessment shall accurately characterize the noise environment, including frequency, intensity, and duration.

2.1.2 Problem Identification: Identify specific noise sources contributing to excessive noise levels in the designated areas.

2.1.3 Solution Development: Develop a range of potential solutions to reduce noise levels to acceptable limits within the defined areas. Solutions should consider various approaches, including but not limited to:

- Modifying SL Exhaust Ducts Without negatively impacting the operation or capabilities of SL2 and SL3
- Installing physical barriers or structures.
- Implementing administrative controls (if appropriate and feasible).
- Investigating alternative noise reduction technologies.

2.1.4 Feasibility Analysis: Evaluate the feasibility (technical, economic, and operational) of each proposed solution. This includes:

- Cost estimations for implementation and maintenance.
- Impact on operations and workflows.

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- Potential for noise level reduction.
 - Installation timelines and requirements.
- 2.1.5 Recommended Solutions: Recommend the most effective and feasible solutions, supported by clear justification and detailed implementation plans.
- 2.1.6 Reporting: Prepare a comprehensive report documenting the study methodology, findings, proposed solutions, feasibility analysis, and recommended actions.

2.2. Milestones

Planned, significant events scheduled to occur at specific times in performance of the SOW. Milestones shall include, at a minimum

- Project Kickoff Meeting
- Business Process Review for the SL Duct Noise Reduction
- Detailed Project Plan Approved
- Site Access and Logistics Approved
- Baseline Noise Measurements Completed
- Noise Source Identification Completed
- Preliminary Solution Alternatives Identified
- SL Exhaust Duct Modification Options Defined
- Feasibility Analysis of Solutions Completed
- Draft Report Submitted for Review
- Final Report Approved and Delivered

2.3. Data Deliverables

Tangible or intangible data items produced as a result of the work performed, and their anticipated delivery date, method of delivery, format, and quantity.

Data Deliverables shall (at a minimum) include the following:

- Business Process Review
 - Project Management Plan
 - Cost, Schedule, and Performance Tracking
 - Schedule delivered in native file format
 - Subcontracting Plan
 - Staffing plan
 - Work Breakdown Structure (WBS)
 - Resource-loaded Integrated Master Schedule

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- Pricing Considerations for Scrap Metal
- Risk Management Plan
 - Identify risk and mitigation strategies
- Study Proposal Deliverables
 - Sampling Strategy: A detailed sampling strategy document, including:
 - A map clearly identifying all locations where measurements will be taken.
 - A justification for the selection of each location, detailing the rationale and criteria used in its determination.
 - Measurement Protocols: A comprehensive description of measurement protocols, including:
 - Precise timing and duration of each measurement.
 - A clear description of the data to be collected, including data types, units of measurement, and expected data format (e.g., CSV, Excel, specific database schema). Provide examples of what the data will look like.
 - Measurement Uncertainty Analysis: A thorough analysis of measurement uncertainty, including:
 - Identification of potential sources of error.
 - Specific procedures to address the impact of temperature variations and other environmental factors (e.g., humidity, wind speed) on measurement accuracy. This will include details on equipment used to monitor and mitigate these factors.
 - Noise Source Identification/Classification Strategy: A defined strategy for identifying and classifying noise sources that may affect the measurements, including methods for distinguishing between relevant and irrelevant noise.
 - Modeling Methods: A detailed description of the modeling methods to be employed.
 - Validation Methods: A description of the methods to be used to validate the accuracy and reliability of the measurements and the models.
 - Final Report: A comprehensive final report that includes:
 - A summary of the project objectives and methodology.
 - All raw and processed data collected during the project.
 - A detailed analysis of the results, including comparisons of predicted performance.
 - A discussion of any limitations or uncertainties in the data or analysis.

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- All supporting documentation, including calibration records and software code (if applicable).
- Measurement Tolerances: A clear description of the acceptable measurement tolerances for all parameters being measured.
- Calibration Procedures: A detailed description of all calibration procedures
- Regulatory Compliance Documentation: Documentation demonstrating compliance with all applicable regulatory requirements and standards.
- Codes/Standards Evaluation: A comprehensive evaluation against the list of applicable codes and standards, demonstrating adherence and highlighting any areas of non-compliance, along with proposed mitigation strategies.
- Proof of Concept
 - Conceptual Design
 - The vendor shall provide an acceptable overview of the following variables for how their company plans to work with the Government to reach an approved conceptual design in an expedient manner, if awarded the prototype agreement.
 - Clear identification of Equipment Models and their specifications.
 - Preliminary structural analysis to verify feasibility of design
 - Availability of detailed Equipment Specifications defining the equipment's design and operating parameters.
 - Comprehensive Non-Destructive Examination Results to ensure equipment integrity and safety.
 - Certified Pressure Test Documentation ensuring equipment meets pressure-containment standards.
 - Proactive Life Cycle Support Plan to ensure long-term maintenance and availability.
 - Detailed Manufacturing Plan providing traceability of equipment components and manufacturing processes.
 - Robust Maintenance and Reliability Documents outlining maintenance schedules, procedures, and reliability analyses.
 - Provide list of possible hazards and assess how they can be minimized or eliminated.
 - Maintenance and Reliability Documents

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2.4. Data Rights

The Government shall receive Unlimited Data Rights to all Contractor's technical data and drawings (i.e., data deliverables) produced in performance of the SOW.

2.5. Training:

The Government shall receive on-site orientation and training for all new equipment or systems. Training will be recorded by the Government for future use.

3. Quality Control Plan (QCP)

The contract shall define the development and maintenance of a quality control plan to ensure the requirements are met in accordance with the SOW. This shall include the development and implementation of procedures to meet milestones and to ensure appropriate corrective action is taken when milestones are not met.

4. Program Management Objective

Allow offerors the maximum flexibility to innovatively manage schedule, performance, risks, warranties, subcontracts, and data so that it satisfies the user's performance requirements (see SOO Attachment 1). Clear government visibility into the schedule, performance, and risks shall be maintained.

Attachment:

1. SOO Attachment 1 – Performance Parameters - Distro D

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